



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Production Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – II)

Class: Final Year	Subject: Tools for Six Sigma Quality	Subject Code: PCC-PE403
Date: 23.03.2026	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Solve the following questions (Any Four).	M	CO	BT
1	What are the key objectives of the Measure Phase in Six Sigma, and why is it important to measure the current process before implementing improvements?	05	CO3	BT1
2	Describe and explain the different graphical tools used in the Measure Phase to analyze data. How can they help in problem-solving?	05	CO3	BT2
3	What are DPO (Defects Per Opportunity) and DPMO (Defects Per Million Opportunities)? How are they used to evaluate process performance in Six Sigma?	05	CO3	BT1
4	What are SIPOC diagrams, and how are they used in the Measure Phase of Six Sigma? Describe each component of a SIPOC diagram?	05	CO3	BT2
5	Explain the concept of Process Capability and how it is measured. What does a high process capability indicate?	05	CO3	BT2

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Mechanical Engineering
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: B. Tech Course Name: Quality and Reliability Engineering Course Code: PCC – ME 412

Date: 23/03/2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

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Q. N.		M	CO	BT
1	What are the types of control charts? Explain $\bar{X}$ chart in detail.	05	CO1	BT1
2	Explain step by step how to construct $\bar{X}$ and R chart.	05	CO2	BT2
3	The thickness of the magnetic coating on audio tapes is an important characteristic. Random samples of size 4 are selected, and the thickness is measured using an optical instrument. Table (refer it which is attached with it) shows the mean $\bar{x}$ and standard deviation s for 20 samples. The specifications are 38 ± 4.5 micrometers. If a coating thickness is less than the specifications call for, that tape can be used for a different purpose by running it through another coating operation. Find trial control limit for $\bar{X}$ and s -chart.	05	CO2	BT3
4	What are the benefits and limitations of attribute chart.	05	CO1	BT2

== ALL THE BEST ==

Sample	Sample Mean, $\bar{X}$	Sample Standard Deviation, s	Sample	Sample Mean, $\bar{X}$	Sample Standard Deviation, s
1	36.4	4.6	11	36.7	5.3
2	35.8	3.7	12	35.2	3.5
3	37.3	5.2	13	38.8	4.7
4	33.9	4.3	14	39.0	5.6
5	37.8	4.4	15	35.5	5.0
6	36.1	3.9	16	37.1	4.1
7	38.6	5.0	17	38.3	5.6
8	39.4	6.1	18	39.2	4.8
9	34.4	4.1	19	36.8	4.7
10	39.5	5.8	20	37.7	5.4



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.

Department of Mechanical Engineering

Academic Year: 2025-2026 [SEM. – 2]

In-Semester Examination - 2

Class: Final Year, B.Tech.

Subject: Operations Research

Subject Code: HMC-ME402

Date: 28/03/2026

Time: 14.30-15.15

Max. Marks: 20

NB:

- I. This paper contains One page (*Three Questions*). Figures to right indicate full marks. All Questions are Compulsory.
- II. Outline your solution in clear, sequential steps.

Q.No.	Questions	Marks	CO	BT
1.	In an Industry, five different machines can do any of the five jobs, with different profit resulting from each assignment. Find the assignment that maximizes the total profit & amount of total profit. [Hint: Maximization Problem]	10	CO1, 2	BT3, 4

		<u>Machines</u>				
		A	B	C	D	E
<u>Jobs</u>	I	30	37	40	28	40
	II	40	24	27	21	36
	III	40	32	33	30	35
	IV	25	38	40	36	36
	V	29	62	40	34	39

2. What does a *critical path* signify? In what ways all its activities are particularly important? (02 Marks)

A Project has the following activities & other characteristics:

Activities	Predecessor	Time Duration (Weeks)		
		Optimistic	Most Likely	Pessimistic
A	-	4	7	16
B	-	1	5	15
C	A	6	12	30
D	A	2	5	8
E	C	5	11	17
F	D	3	6	15
G	B	3	9	27
H	E, F	1	4	7
I	G	4	19	28

Draw the Project Network, Identify the *critical path* and find the *expected project completion time*. Also find the probability of completion of project within 40 weeks. (08 Marks)
